

A stylized blue manta ray graphic is positioned on the left side of the page. The ray is depicted in a dark blue color with lighter blue, mottled patterns on its body, suggesting a natural skin texture. It is oriented diagonally, with its head pointing towards the top left and its tail extending towards the bottom right. The pectoral fins are spread out, and the tail is slightly curved.

Manta MK2

Contents

1. Parts Preparation
2. Assembly
3. Direct Drives
4. BL Touch
5. Gallery
6. Additional Notes

BOM

1. M3 Thermal Heat sets\Inserts OD 5.0 x 5mm long (4.25mm ID hole specified in plastic) 25 - 30 required
2. 1 x 3010 Hot end Cooling Fan 24 volt.
3. 2 x 4010 or 5015 part Cooling Fans 24 volt.
4. Various M3\M2 Nuts & Bolts

2 x M3 x 25mm Hex Bolts
4 x M3 x 20mm Hex Bolts
4 x M3 x 12mm Hex Bolts
3 x M3 x 6mm Hex Bolts
17 - 22 x M3 x 8mm Hex Bolts
4 x M3 x Plain Nuts (2 x used for Stock spacers)
2 x M2 x 10mm Hex Bolts
2 x M2 x 8mm Hex Bolts

5. 4 x small Cable Ties

6. Heat shrink, soldering, insulation tape for safe finishing of cables & connection of cables.

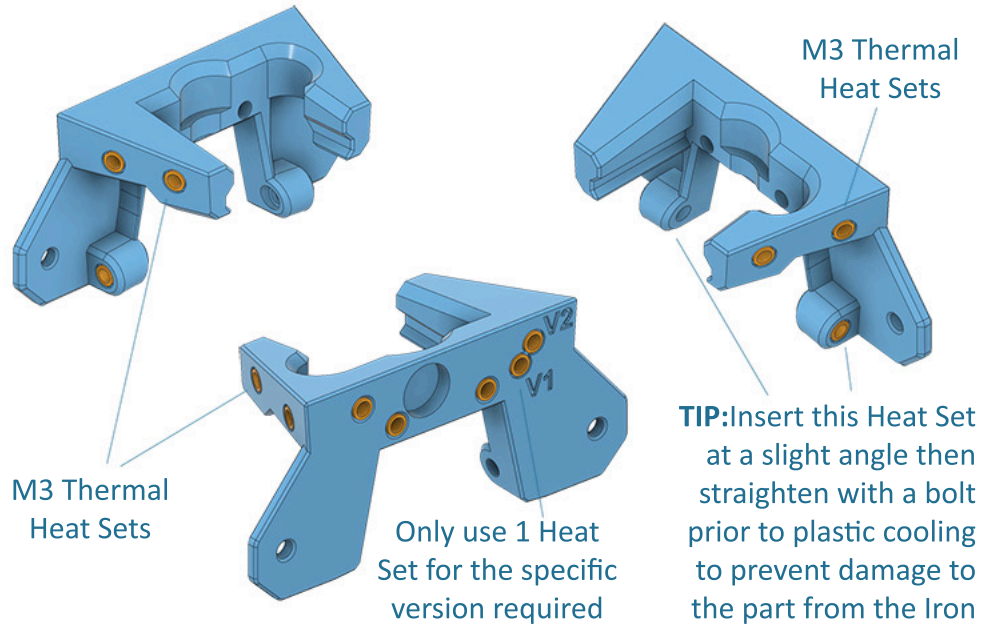
7. Short Length of Bowden tube for Direct Drive Configuration.

Useful Tools

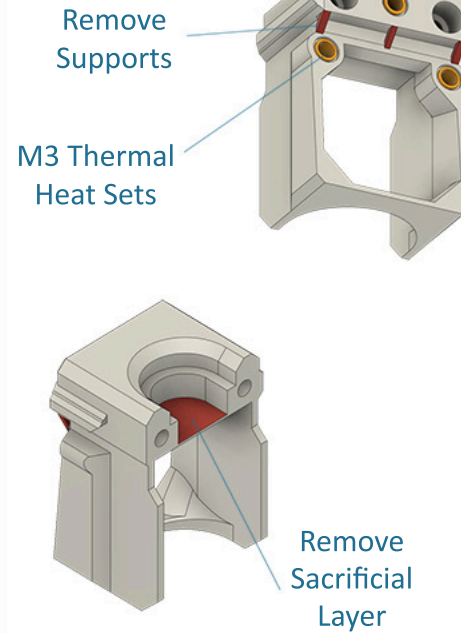
1. M3 tap for cleaning plastic out of Heat set holes
2. M3 nut wrench
3. Cable stripping tool

Part Preparation 1

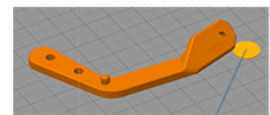
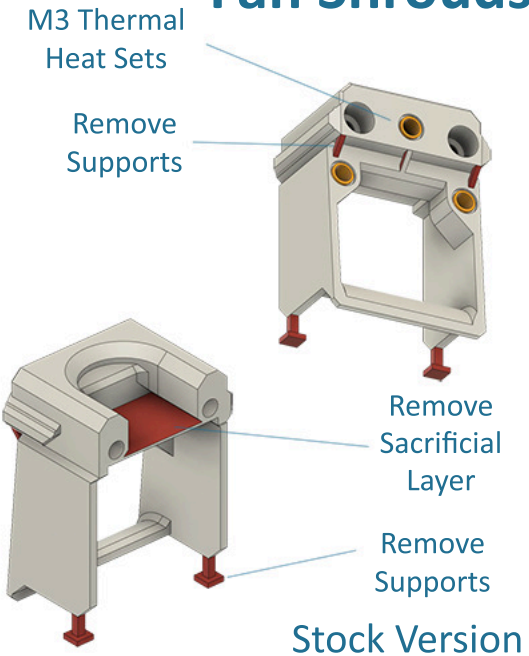
Duct Base



V6 Version

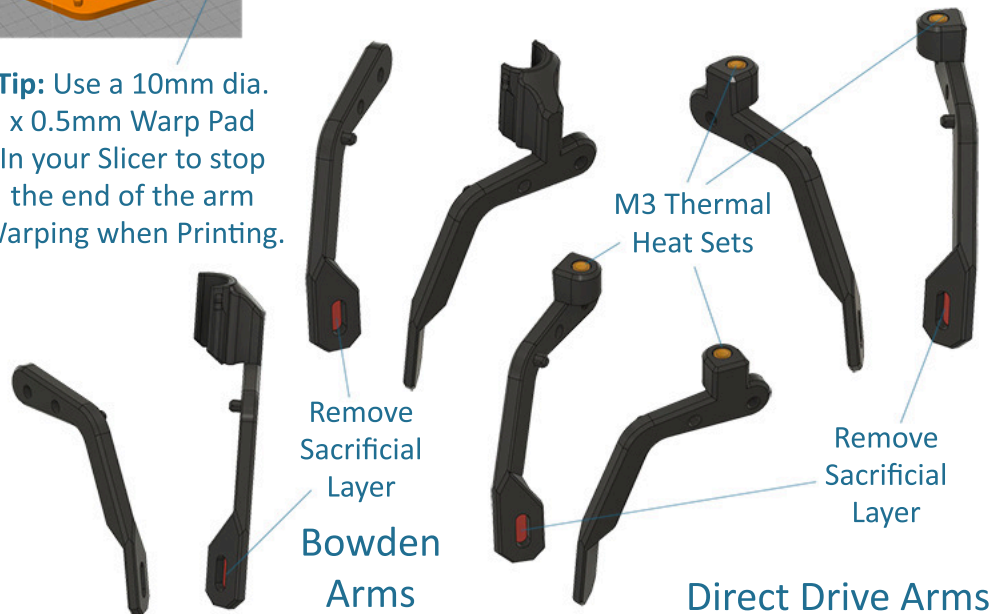


Fan Shrouds

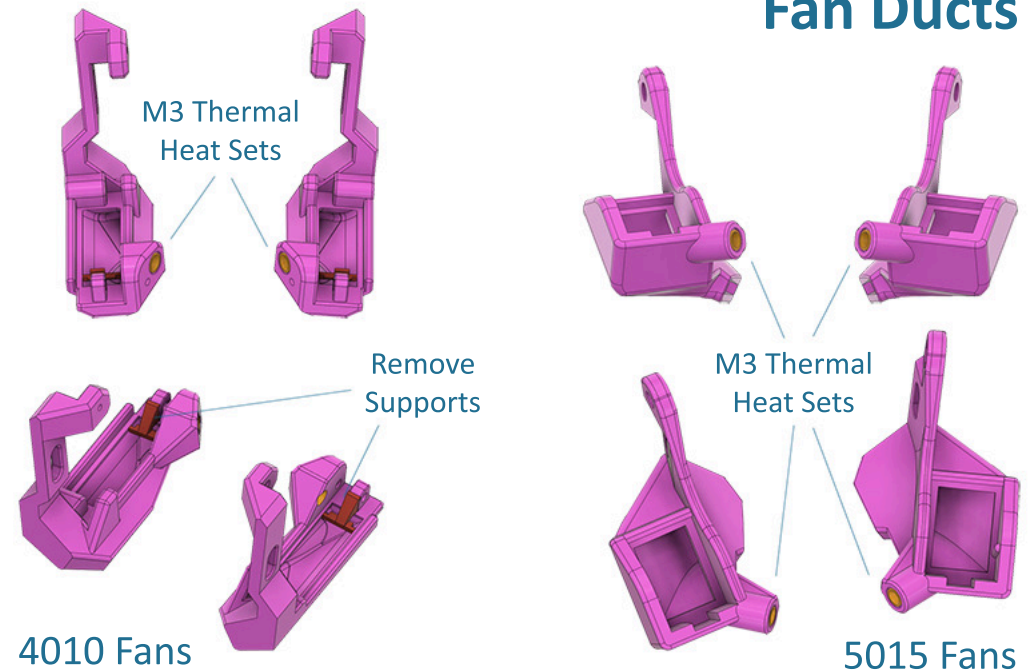


Tip: Use a 10mm dia. x 0.5mm Warp Pad In your Slicer to stop the end of the arm Warping when Printing.

Support Arms

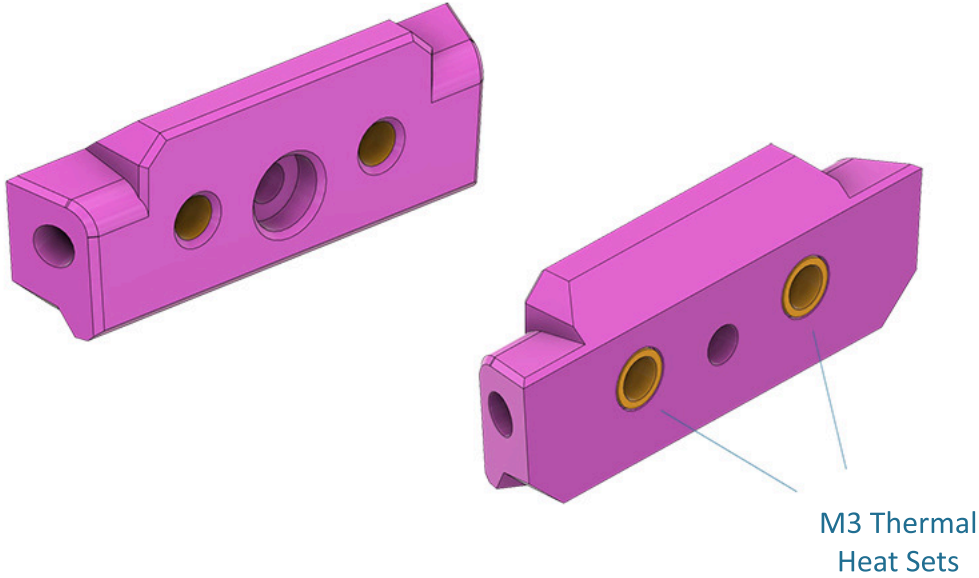


Fan Ducts



Part Preparation 2

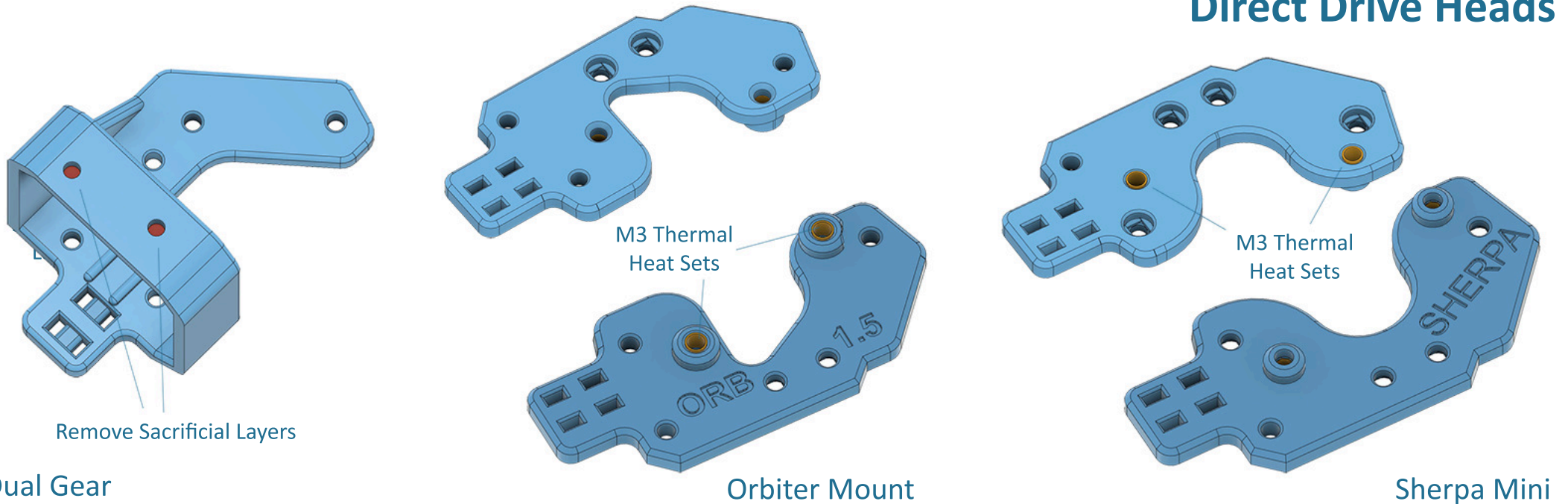
BL Touch Bracket



Rear Mount

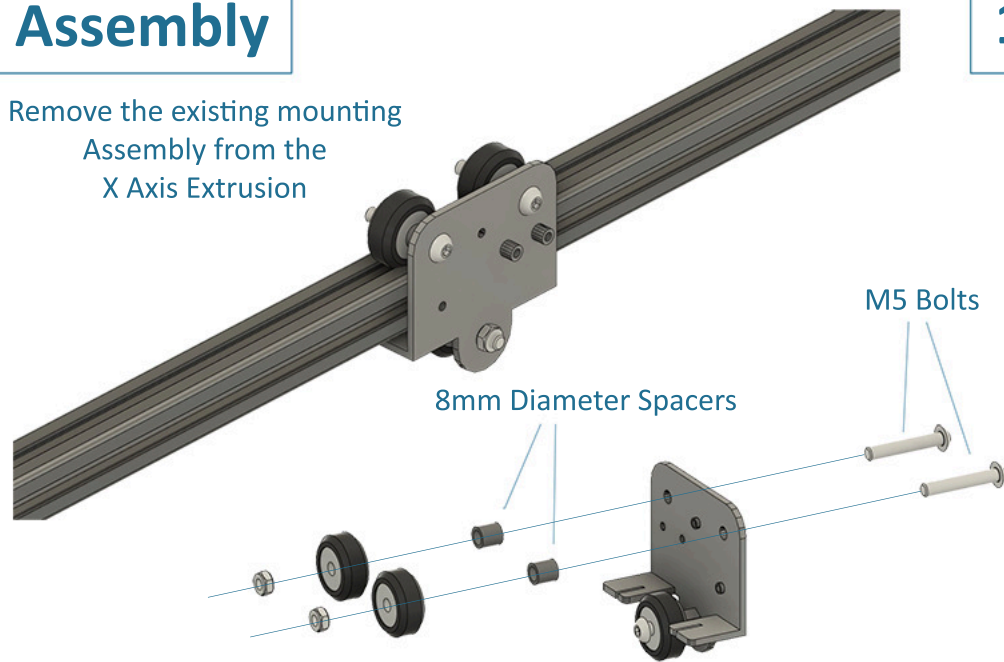


Direct Drive Heads



Assembly

Remove the existing mounting Assembly from the X Axis Extrusion



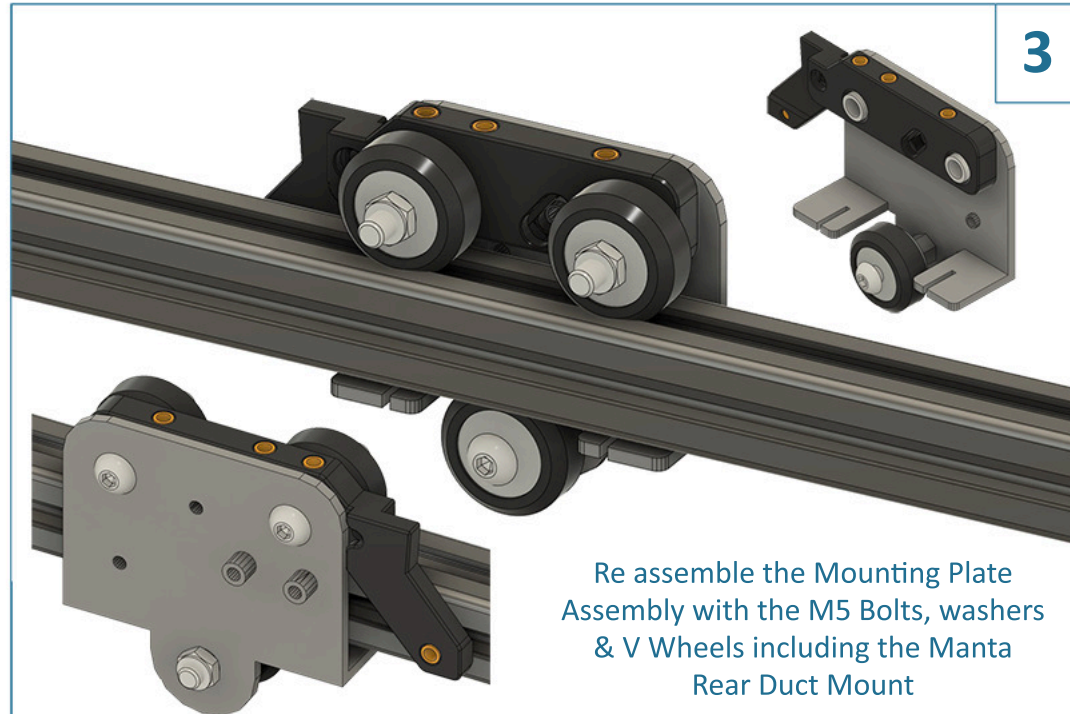
1

8mm Spacers



Take the two 8mm spacers from the stock assembly and insert them into the Rear Mount with them flush to the inner face and slightly protruding from the outer face.

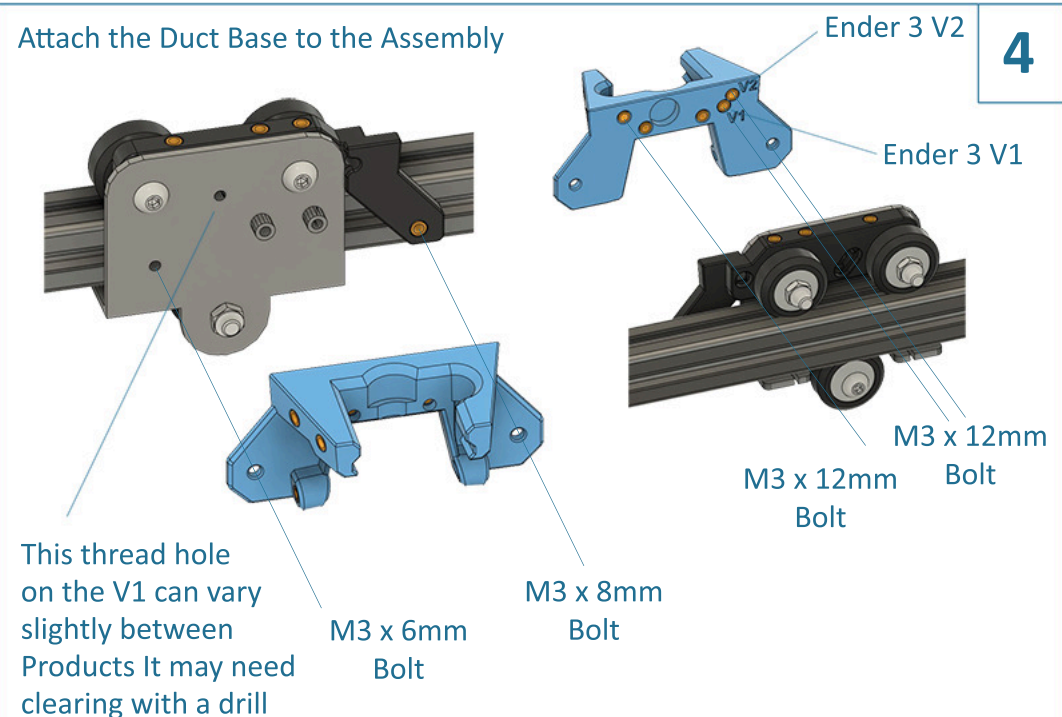
2



3

Re assemble the Mounting Plate Assembly with the M5 Bolts, washers & V Wheels including the Manta Rear Duct Mount

Attach the Duct Base to the Assembly



4

Ender 3 V2

Ender 3 V1

M3 x 12mm Bolt
M3 x 12mm Bolt

M3 x 8mm Bolt

M3 x 6mm Bolt

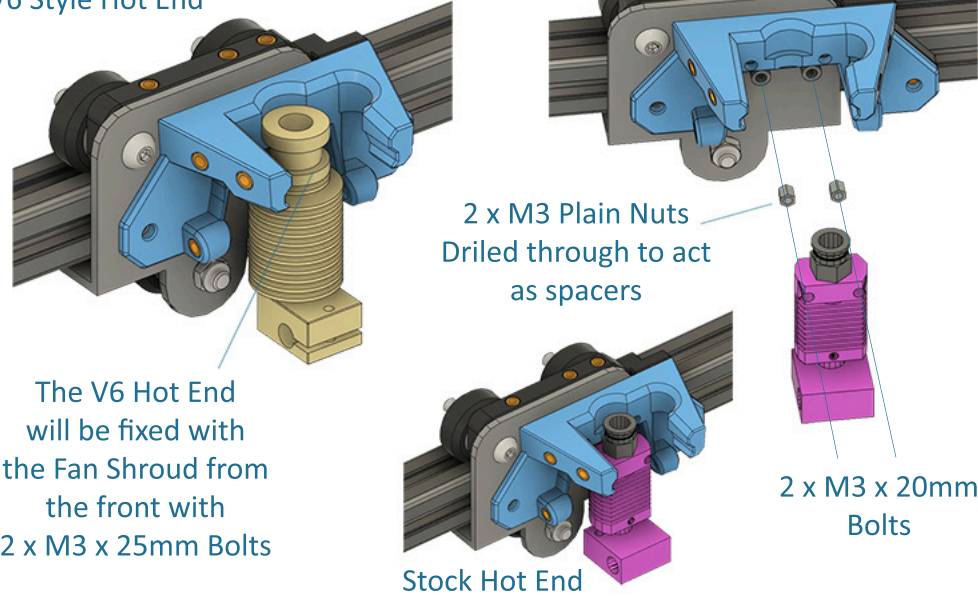
This thread hole on the V1 can vary slightly between Products It may need clearing with a drill

Assembly

Prepare the Hot End for Assembly Stock Hot End

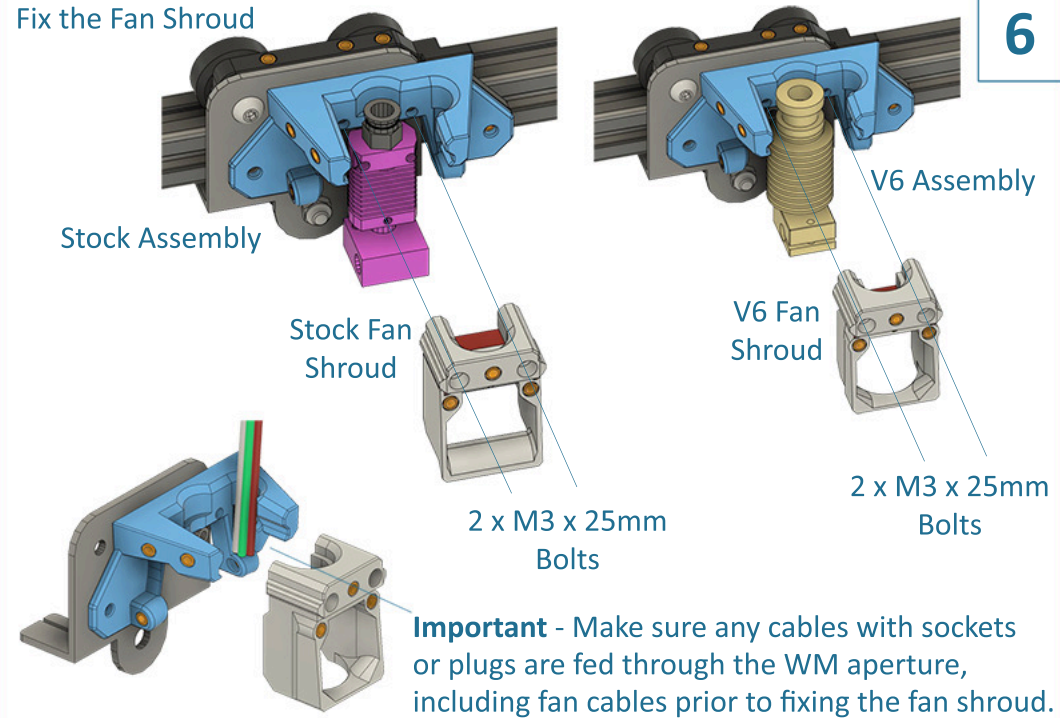
5

V6 Style Hot End



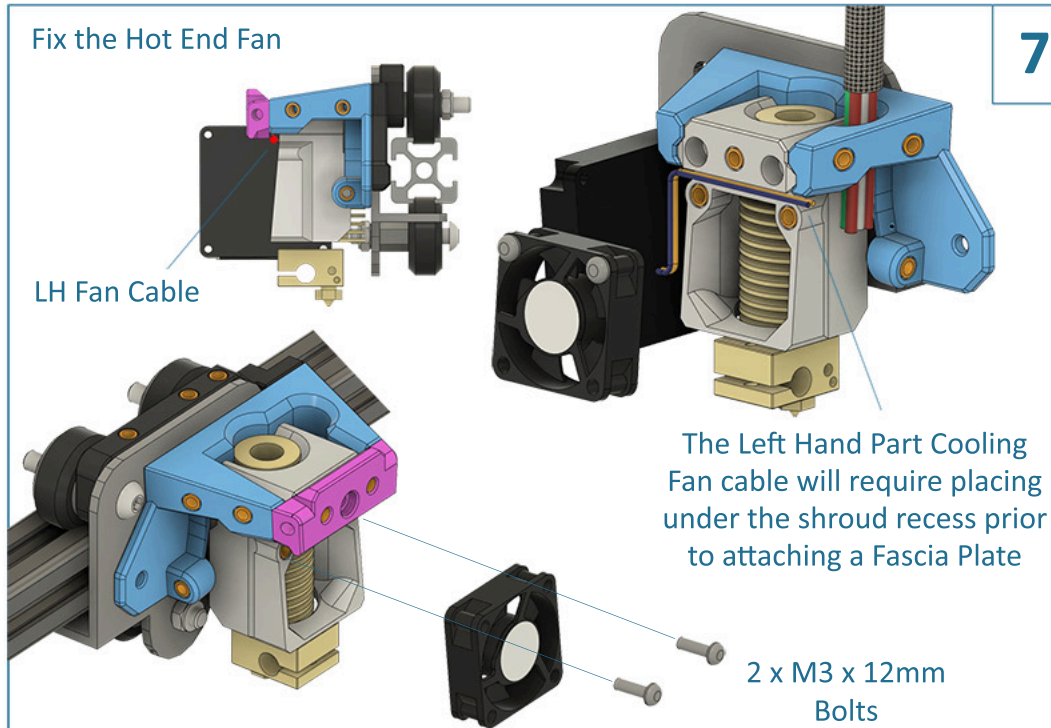
Fix the Fan Shroud

6



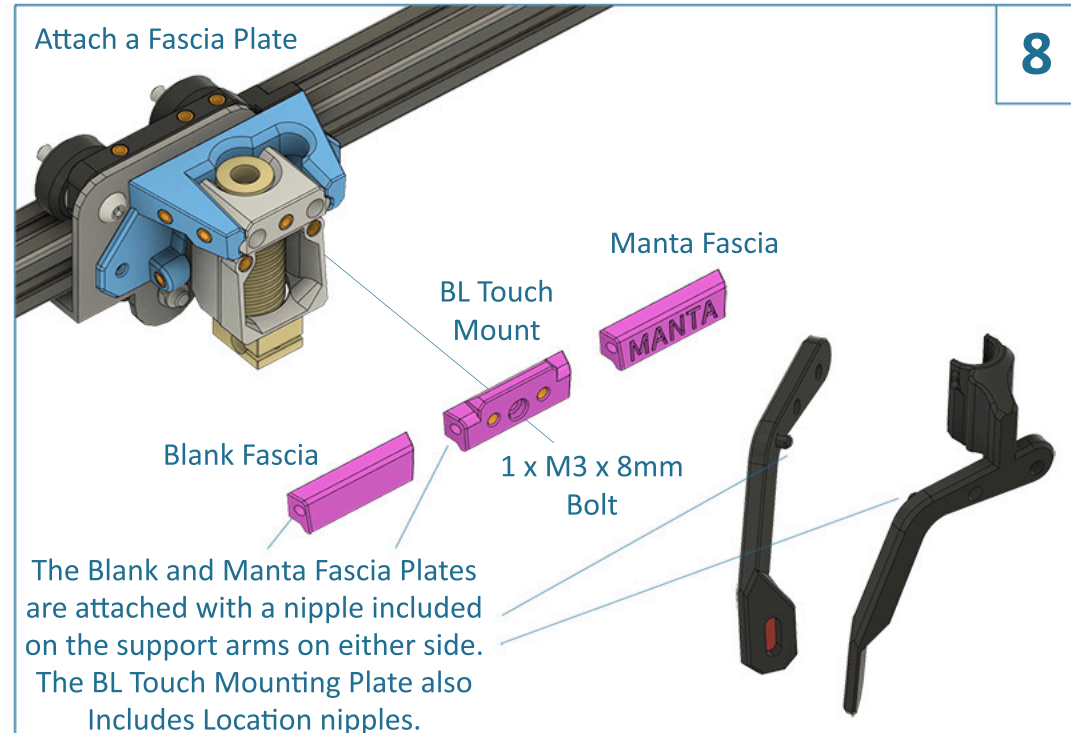
Fix the Hot End Fan

7



Attach a Fascia Plate

8

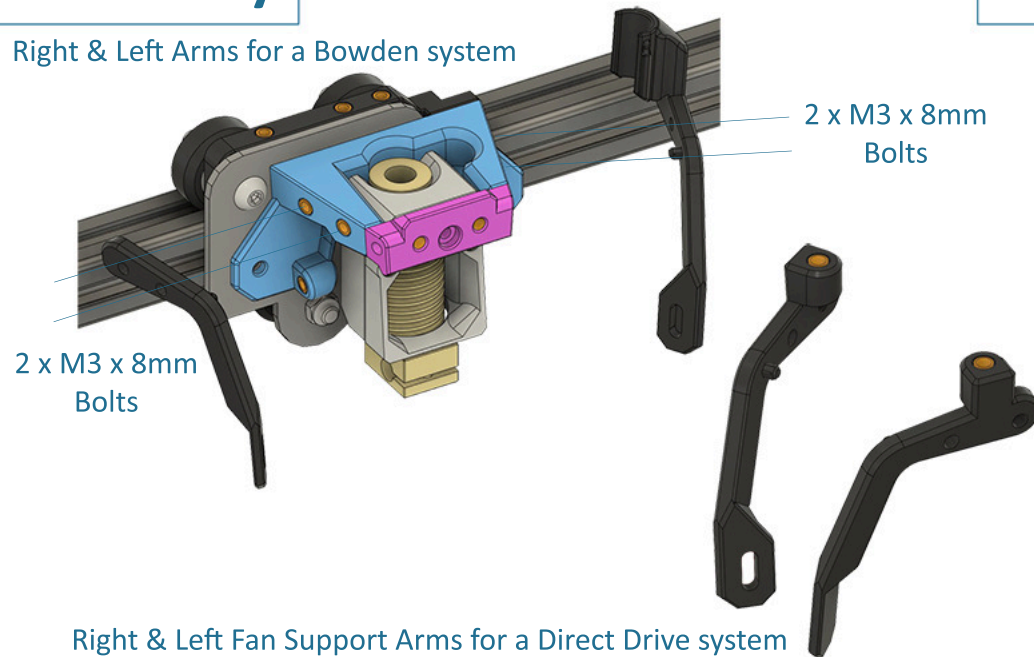


Assembly

Attach the Fan Support Arms

9

Right & Left Arms for a Bowden system



Attach the Fans to the Fan Ducts

4010 Fans

2 x M2 x 8mm Bolts

2 x M2 x 10mm Bolts

2 x M3 x 20mm Bolts

5015 Fans

2 x M3 Plain Nuts

Duct Alignment from Build Plate

10

4010 Fans
2mm

5015 Fans
3mm

Tip: Use the provided Spacer\Helper to set the Ducts off the Build Plate by Inserting under with the nozzle touching at the home position



Attach the Fan Ducts to the Duct Base - Left Hand First

11



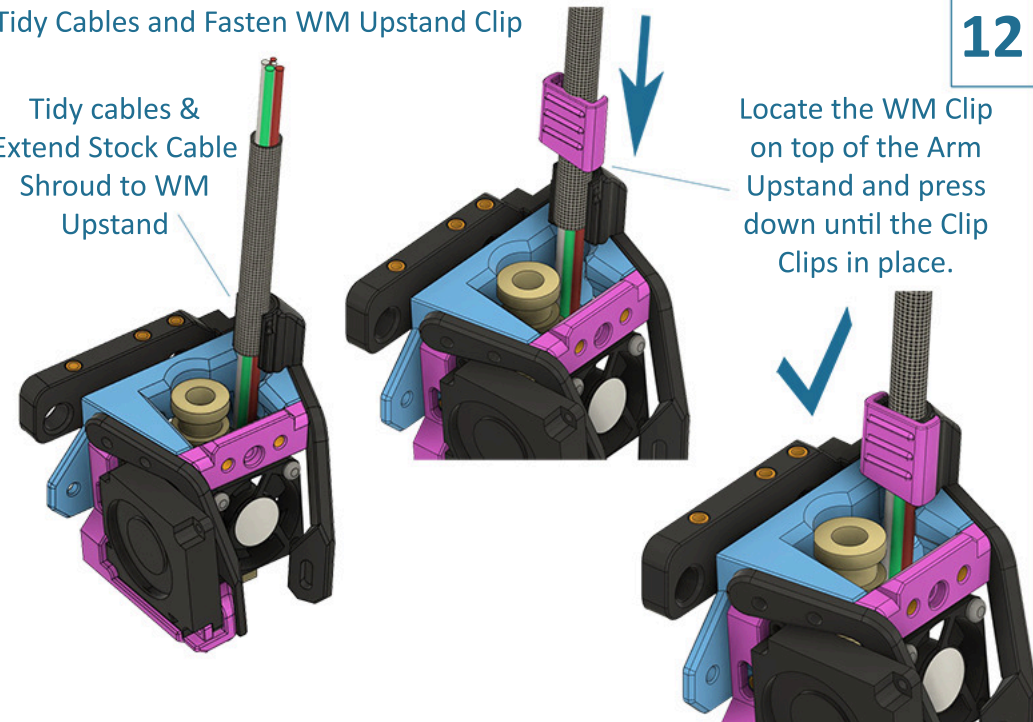
Please Note: You will need to splice the cable for the existing Part Cooling Fan and connect two fans to the remaining lead to supply power to both fans.

Tidy Cables and Fasten WM Upstand Clip

12

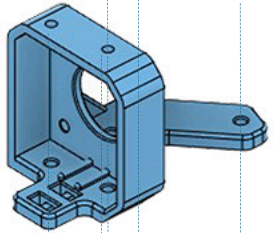
Tidy cables & Extend Stock Cable Shroud to WM Upstand

Locate the WM Clip on top of the Arm Upstand and press down until the Clip Clips in place.

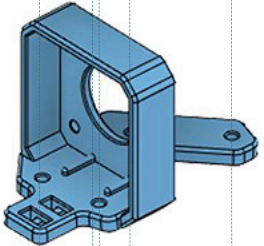


Direct Drives

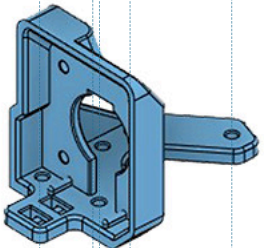
5 x M3 x 8mm Bolts



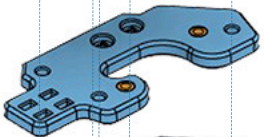
Dual Gear
+ Stock motor
Fixings



Stock
+ Stock motor
Fixings

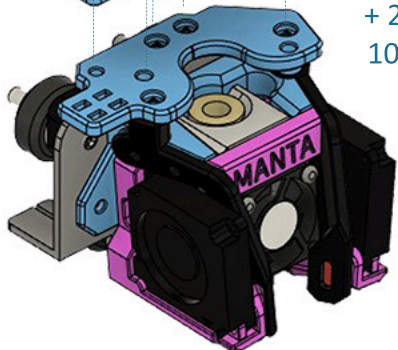


BMG
+ Stock motor
Fixings



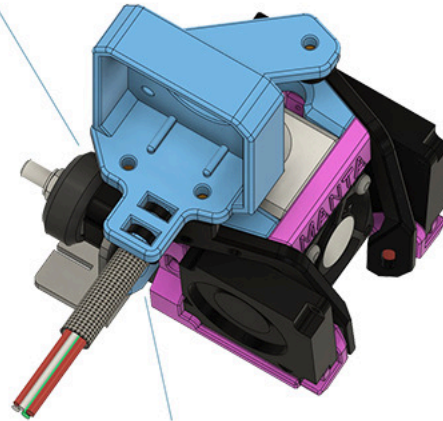
Orbiter 1.5
+ 2 x M3 x
10-12mm

Sherpa Mini
+ 2 x M3 x
10-12mm



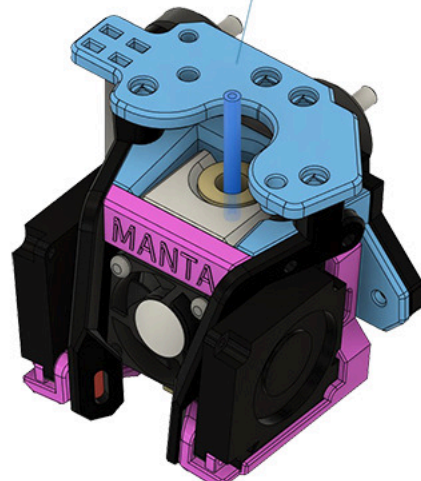
Tip: Insert cable ties before
attaching stepper motor

1

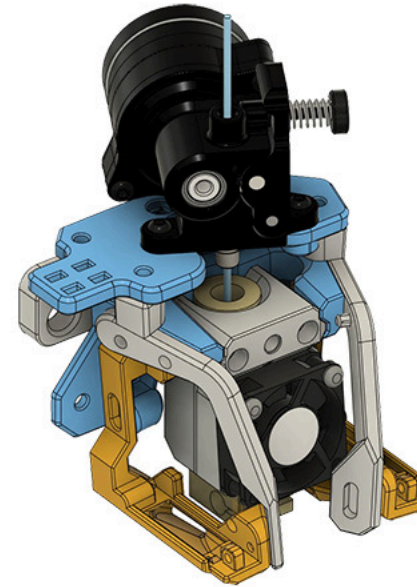


Cable Tie Detail on all
Tool Heads to
aid Wire Management

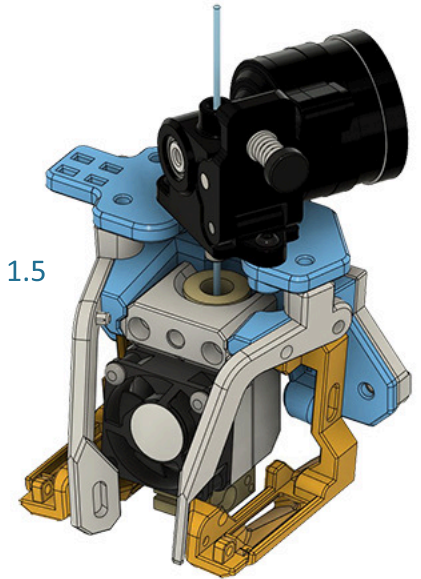
A small Length of Bowden
Tube is required for the
Direct Drive assembly.
This will differ in length
to each type of Drive Head



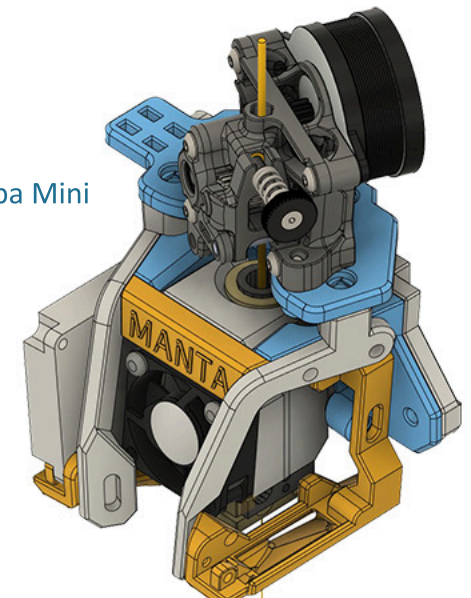
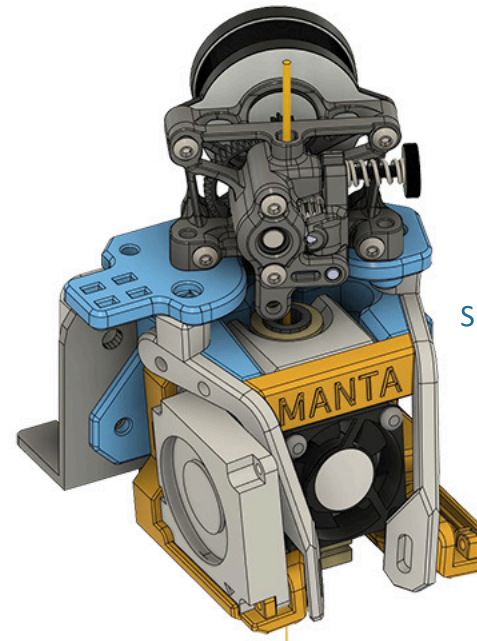
2



Orbiter 1.5



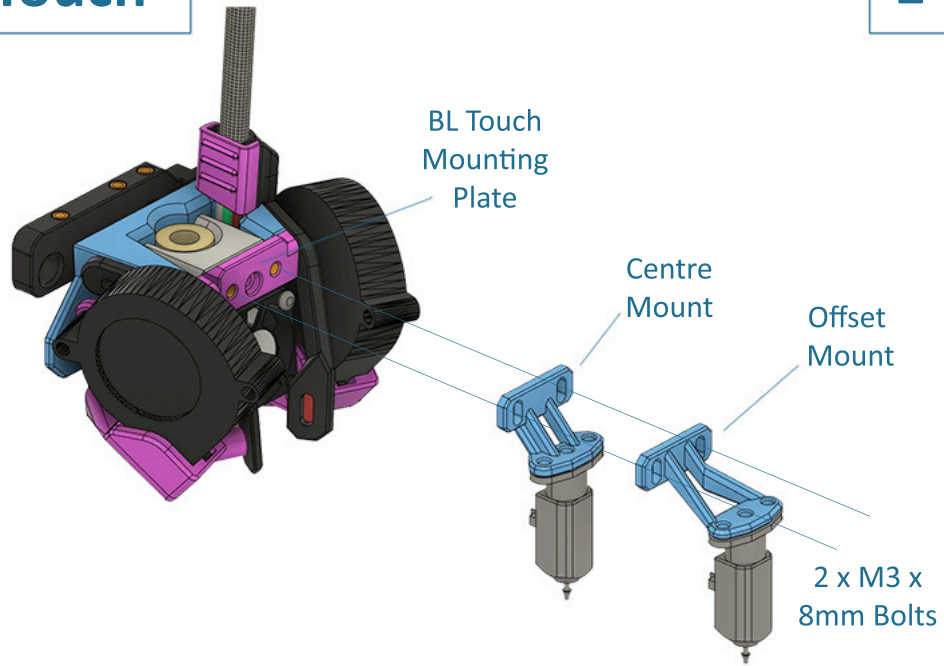
Sherpa Mini



BL Touch

BL Touch Mounts are Work In Progress

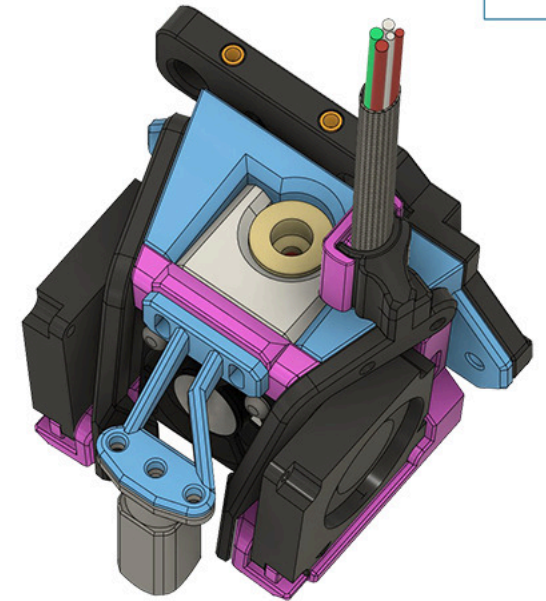
1



Centre Mount

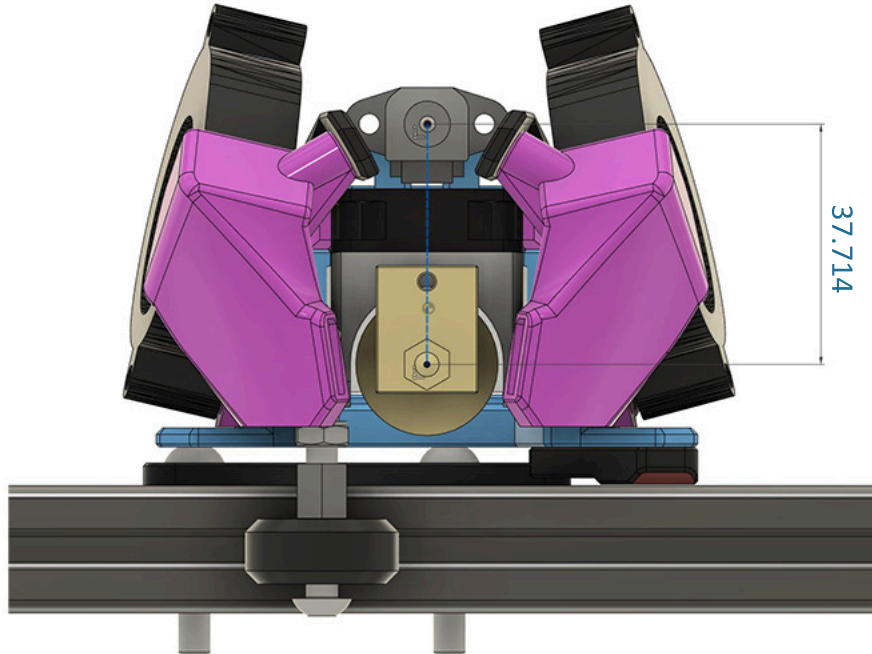


Offset Mount



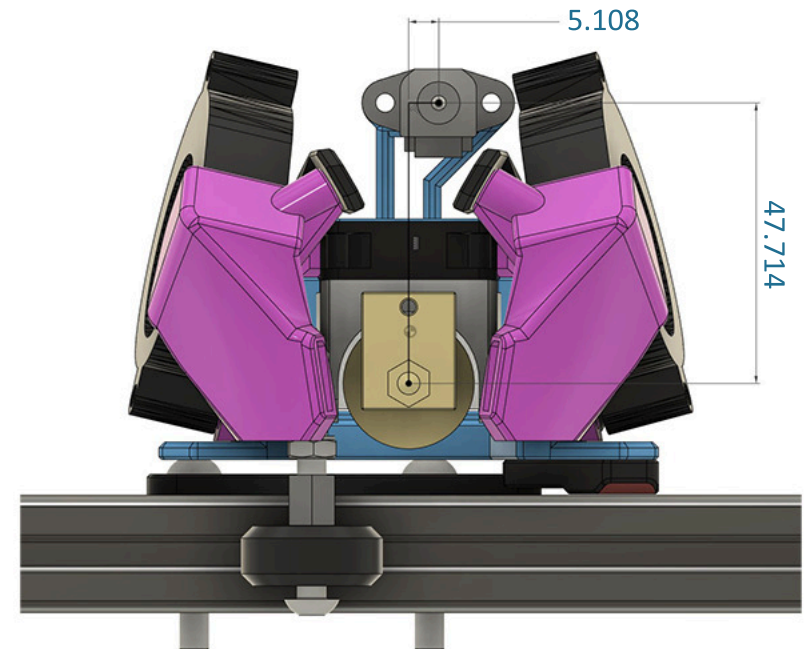
Offsets - Centre Mount

3



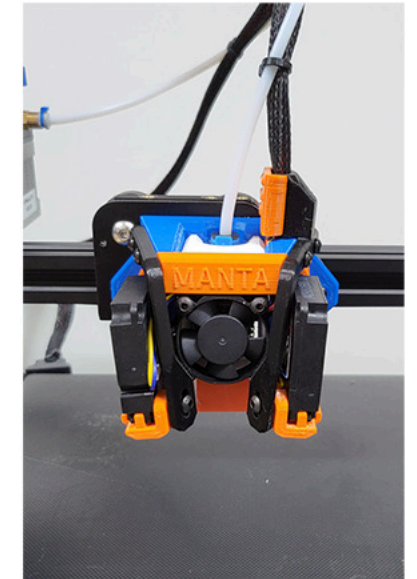
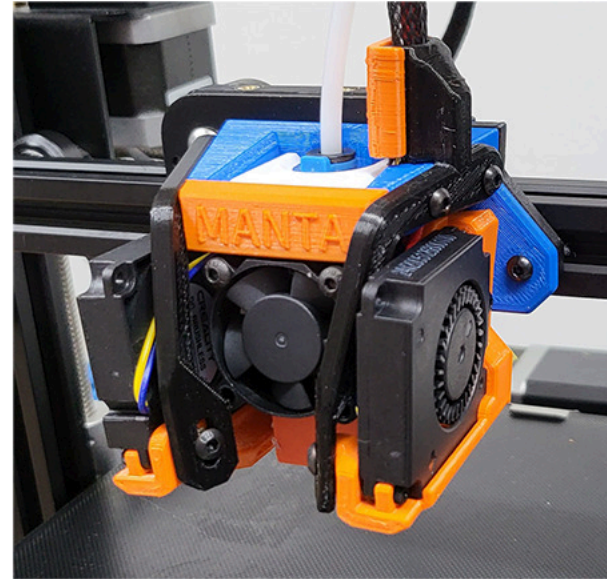
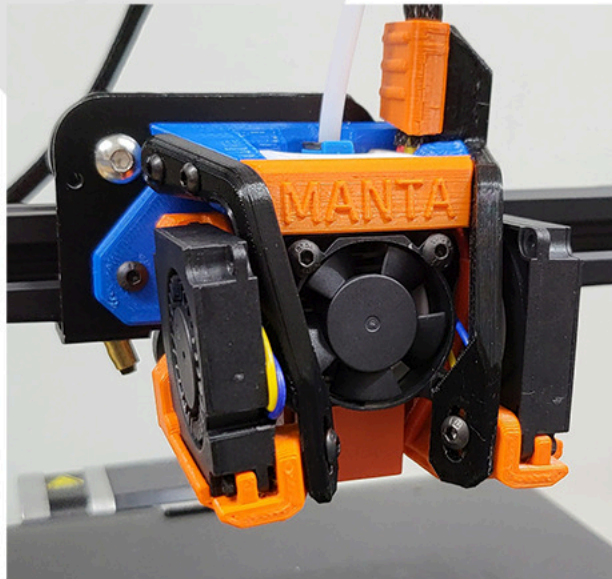
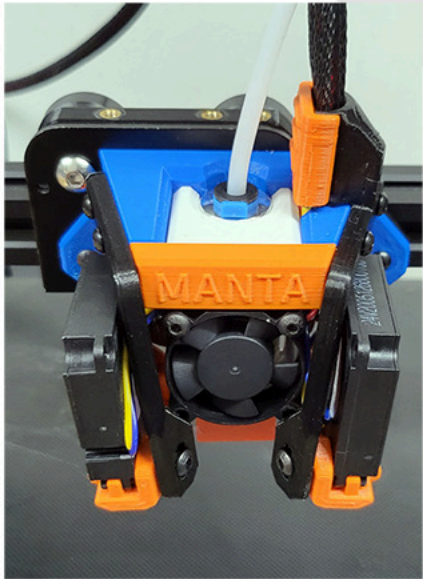
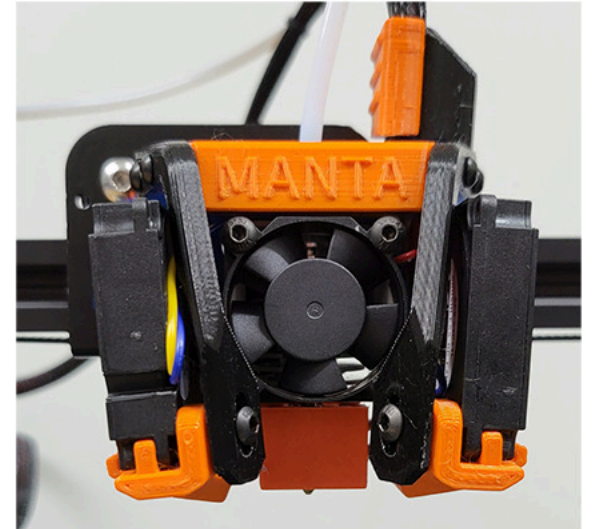
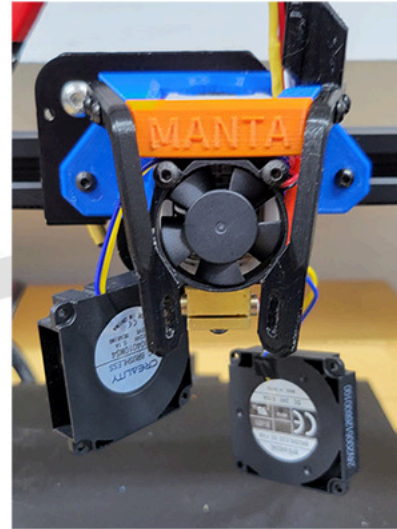
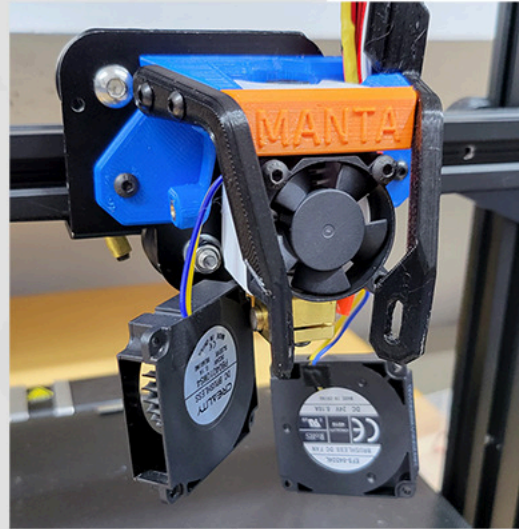
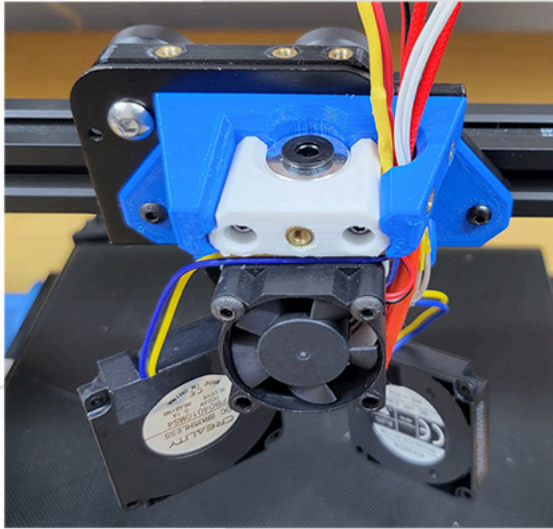
Offsets - Offset Mount

4



Gallery

Manta MK2 - Ender 3 V2 - V6 Hot End with 4010 dual fans - Bowden Configuration



Notes

